

5. Queue build-up is possible only in the Network Layer of the computer network stack. *

Mark only one oval.

- ☒ No it can get build up in any layer
- ☐ No Only in network and application layer
- ☐ No only in data link layer in addition
- ☐ None of the above

6. A study of the Markov process is needed to understand queueing theory because *

Mark only one oval.

- ☐ We see the dynamics as a Markov Process
- ☐ A queue can be analyzed as a Markov Chain
- ☐ A queue is a Markov chain where all states must be reachable from each other
- ☒ All of the above

7. A Markov chain means *

Mark only one oval.

- ☐ All states are reachable from each other in a single transition
- ☒ Every pair of the states is reachable from each other
- ☐ At least one state needs to be transient
- ☐ All of the above

8. Service rate in a queue in a typical Network Layer in the computer network stack depends on *

Mark only one oval.

- ☐ How fast the cables attached with the router can be repaired
- ☐ How fast the processor of the router can process the packets
- ☒ How fast the outgoing link of router can transmit the data
- ☐ None of the above

9. *

Consider an M/M/1 queue modeled as a birth–death process with:

- Arrival rate = n
- Service rate = m
- Infinite buffer capacity
- $m > n$ (stable system)

What is the steady-state probability that exactly **one customer** is present in the system?

- (a) $\frac{m(m-n)}{n^2}$
- (b) $\frac{m(m-n)}{n^3}$
- (c) $\frac{m(m-n)}{m^2}$
- (d) $\frac{n(m-n)}{m^2}$

Mark only one oval.

- ☐ Option (a)
- ☐ Option (b)
- ☐ Option (c)
- ☒ Option (d)

10. *

Consider an M/M/1 queue with:

- Arrival rate = n
- Service rate = m
- Maximum system capacity $p = 1$

What is the steady-state probability that the system contains exactly **one** customer?

(a) $m + n$

(b) $\frac{m}{m + n}$

(c) $\frac{n}{m + n}$

(d) $1 - \frac{n}{m}$

Mark only one oval.

☐ Option (a)

☐ Option (b)

☒ Option (c)

☐ Option (d)

11. A rate transition diagram and a state transition diagram are - *

Mark only one oval.

☐ Yields same results

☐ State transition diagram is more detailed

☐ Its quicker to deal with a rate transition diagram

☒ All of the above

12. ALOHA protocol was invented because *

Mark only one oval.

- ☐ CSMA was not working
- ☒ Wired communication was difficult
- ☐ Long distance wireless communication were error prone
- ☐ Inter-island distance heterogeneity

13. Consider **slotted** ALOHA. If packet sizes differ, will the protocol's performance be affected? [performance implies mainly throughput here] *

Mark only one oval.

- ☒ No, there will be no effect on either throughput or success probability
- ☐ Yes, there will be effect on throughput but not on success probability
- ☐ Yes, there will be effect on success probability but not on throughput
- ☐ Yes, there will be effect on both success probability and throughput

14. If the transmission of **packet p** from **node x** partially overlaps with the transmission of **packet q** from **node y** [i.e., before the transmission of **p** from **x** ends, **q** from **y** starts], then the two packets will always collide at the destination receiver. *

Mark only one oval.

- ☐ No this is never possible
- ☐ Yes, this is always true
- ☒ Sometime it may be false
- ☐ None of the above

15. Consider a scenario with 10 nodes. The distances between the nodes are not negligible, hence RF waves take some finite time to reach from the source to the destination [similar to the islands of Hawai'i!]. Consider the following two statements. *

S1: If a node senses a channel and finds it busy, it does not imply that if it transmits now, there will be a collision

S2: If a node finds the channel idle, it does not imply that if it transmits now, there will be a collision

Mark only one oval.

- ☐ Both S1 and S2 true
- ☒ S1 is true but S2 is false
- ☐ S1 is false but S2 is true
- ☐ Both false

16. Consider **un-slotted** ALOHA. If packet sizes differ, will the protocol's performance be affected? *

Mark only one oval.

- ☐ No, there will be no effect on either throughput or success probability
- ☐ Yes, there will be effect on throughput but not on success probability
- ☐ Yes, there will be effect on success probability but not on throughput
- ☒ Yes, there will be effect on both success probability and throughput

17. Consider the following statements:

*

S1: The data link layer for a short-range, shared wireless medium broadcast network does not need to address reliability issues that much.

S2: The data link layer for a point-to-point wired network does not need to deal with collisions among the data packets.

Which of the following options is correct?

Mark only one oval.

- ☐ S1 is true, S2 is false
- ☐ S1 is false, S2 is true
- ☒ Both S1 and S2 are true
- ☐ Both S1 and S2 are false

18. What is true about ALOHA and CSMA 0-persistent (0-p)? *

Mark only one oval.

- ☐ ALOHA senses channel but CSMA does not
- ☒ CSMA senses channel but ALOHA does not
- ☐ Both senses channel
- ☐ None senses channel

19. Consider CSMA 0-p (0-persistent) and CSMA 1-p (1-persistent). Which one of the following is true?

Mark only one oval.

- ☐ Both are equally good in heavily loaded system
- ☐ CSMA 0-p would be the better in all cases
- ☐ CSMA 0-p is better under light load but CSMA 1-p is better under heavy load
- ☒ None is true

20. *

Let P be the transition probability matrix of a Markov chain and let P^4 denote the 4-step transition probability matrix.

What does $P_{31}^{(4)}$ represent?

- (a) It represents the probability that the chain starts in state 4 and reaches state 1 in 3 steps.
- (b) It represents the probability that the chain moves from state 3 to state 1 in exactly 4 steps.
- (c) It represents the probability that the chain is in state 3 after 4 steps given that it started in state 1.
- (d) It represents the probability that the chain starting from state 3 visits state 1 at least once during the first 4 steps.

Mark only one oval.

☐ a

☒ b

☐ c

☐ d

21. *

Consider the following two 3×3 matrices:

$$P_1 = \begin{bmatrix} 0.5 & 0.3 & 0.2 \\ 0.1 & 0.6 & 0.3 \\ 0.4 & 0.2 & 0.4 \end{bmatrix}$$

$$P_2 = \begin{bmatrix} 0.5 & 0.3 & 0.2 \\ 0.3 & 0.4 & 0.3 \\ 0.2 & 0.3 & 0.5 \end{bmatrix}$$

Which of the following statements is correct?

- (a) P_1 is stochastic but not doubly stochastic, and P_2 is doubly stochastic.
- (b) Both P_1 and P_2 are doubly stochastic.
- (c) P_1 is doubly stochastic and P_2 is only stochastic.
- (d) Neither P_1 nor P_2 is stochastic.

Mark only one oval.

- ☒ Option (a)
- ☐ Option (b)
- ☐ Option (c)
- ☐ Option (d)

22. A queueing system is modeled as a Markov process where the state is defined as the number of packets in the system. Which of the following situations would violate the Markov property?

Mark only one oval.

- ☐ Packet arrivals follow a Poisson process.
- ☐ Service times are exponentially distributed.
- ☒ Service time depends on how long the current packet has already been in service.
- ☐ The next arrival is independent of previous arrivals.

23. In a birth–death Markov process used to model an M/M/1 queue, which of the following is always true?

Mark only one oval.

- ☐ The process can jump from state i to state $i+2$ in one step.
- ☐ Transition probabilities depend on past states.
- ☒ From any state i , the process can only move to state $i-1$, i , or $i+1$.
- ☐ The arrival rate must always be greater than the service rate.

24. In Aloha analysis, the offered load G is assumed to be Poisson distributed even though retransmissions are not truly Poisson.

Why is this assumption made?

Mark only one oval.

- ☐ Because retransmissions are independent by definition.
- ☐ Because the arrival rate λ is always equal to G .
- ☒ To make the mathematical analysis tractable, even though it is not strictly accurate.
- ☐ Because collisions do not affect the distribution of retransmissions.

25. *

Suppose the 4-step transition probability matrix of a Markov chain is given as:

$$P^{(4)} = \begin{bmatrix} 0.30 & 0.45 & 0.25 \\ 0.20 & 0.50 & 0.30 \\ 0.35 & 0.25 & 0.40 \end{bmatrix}$$

What is the probability that the chain **cannot reach state 3 from state 1 in exactly 4 steps**?

- (a) 0.25
- (b) 0.75
- (c) 0.65
- (d) 0.40

Mark only one oval.

- ☐ Option (a)
- ☒ Option (b)
- ☐ Option (c)
- ☐ Option (d)

26. In a Slotted ALOHA system, time is divided into equal slots, and each slot duration is equal to one frame transmission time. *

A node wants to transmit a packet of size **800 bits** over a channel with data rate **400 bits per second**.

What is the **vulnerable time** for this system?

Mark only one oval.

- ☐ 1 second
- ☒ 2 seconds
- ☐ 4 seconds
- ☐ 0.5 seconds

27. t = time at which a transmission begins

Time unit = packet transmission time

Which statement correctly describes the success condition for a transmission attempt in Pure ALOHA?

Mark only one oval.

- ☐ A transmission is successful if no arrivals occur during the same slot.
- ☐ A transmission is successful if no arrivals occur during the interval $(t, t + 1]$
- ☒ A transmission is successful if no arrivals occur during the interval $(t - 1, t + 1]$
- ☐ A transmission is successful if no arrivals occur during the interval $(t - 1, t]$

28. *

Definitions:

- λ = arrival rate of new packets (per time unit)
- G = offered load (total transmission attempts per time unit, including retransmissions)
- S = throughput (successful transmissions per time unit)

For Pure ALOHA:

$$S = Ge^{-2G}$$

Which of the following statements is **correct**?

- (a) If $\lambda = S$, then retransmissions do not occur and therefore $G > \lambda$.
- (b) If collisions increase, then G increases while λ remains unchanged.
- (c) If $G = \lambda$, then throughput S must be equal to λ .
- (d) Increasing λ always increases throughput S .

Mark only one oval.

- ☐ Option (a)
- ☒ Option (b)
- ☐ Option (c)
- ☐ Option (d)

Definitions:

- G = offered load (total transmission attempts per time unit, including retransmissions)
- S = throughput (successful transmissions per time unit)
- Time unit = packet transmission time

Consider Pure ALOHA and Slotted ALOHA protocols.

Which of the following statements is **correct**?

- (a) Slotted ALOHA has higher maximum throughput because its vulnerable period is twice that of Pure ALOHA.
- (b) In Pure ALOHA, the success probability of a transmission attempt is e^{-G} .
- (c) The maximum throughput of Slotted ALOHA occurs at $G = 1$.
- (d) If the offered load G is very large, the throughput of both protocols approaches 1.

Mark only one oval.

☐ Option (a)

☐ Option (b)

☒ Option (c)

☐ Option (d)